

12.580

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Question

Let $\mathbf{V}$ be the vector space of all 3×3 matrices with complex entries over the real field. If

$$\mathbf{W}_1 = \{\mathbf{A} \in \mathbf{V} : \mathbf{A} = \overline{\mathbf{A}}^T\} \quad \text{and} \quad \mathbf{W}_2 = \{\mathbf{A} \in \mathbf{V} : \text{trace}(\mathbf{A}) = 0\},$$

then find $\dim(\mathbf{W}_1 + \mathbf{W}_2)$.

Solution

$$\mathbf{V} = \text{all } 3 \times 3 \text{ complex matrices over } \mathbb{R} \quad (1)$$

For one complex element 2 real parameters are required.

$$\dim_{\mathbb{R}}(\mathbf{V}) = 3 \times 3 \times 2 = 18 \quad (2)$$

$$\mathbf{W}_1 = \{\mathbf{A} \in \mathbf{V} : \mathbf{A} = \overline{\mathbf{A}}^T\} \quad (3)$$

$$\mathbf{A} = \begin{pmatrix} a_{11} & a_{12} & a_{13} \\ \overline{a_{12}} & a_{22} & a_{23} \\ \overline{a_{13}} & \overline{a_{23}} & a_{33} \end{pmatrix} \quad a_{ii} \in \mathbb{R}, \quad a_{ij} \in \mathbb{C}, \quad i < j \quad (4)$$

$$\dim_{\mathbb{R}}(\mathbf{W}_1) = 3 + 3 \times 2 = 9 \quad (5)$$

$$\mathbf{W}_2 = \{\mathbf{A} \in \mathbf{V} : \text{trace}(\mathbf{A}) = 0\} \quad (6)$$

Solution

So one element along diagonal has constraint.

$$\dim_{\mathbb{R}}(\mathbf{W}_2) = ((3 \times 3) - 1) \times 2 = 16 \quad (7)$$

$$\dim_{\mathbb{R}}(\mathbf{W}_1 \cap \mathbf{W}_2) = 9 - 1 = 8 \quad (8)$$

$$\dim(\mathbf{W}_1 + \mathbf{W}_2) = \dim(\mathbf{W}_1) + \dim(\mathbf{W}_2) - \dim(\mathbf{W}_1 \cap \mathbf{W}_2) \quad (9)$$

$$= 9 + 16 - 8 \quad (10)$$

$$= 17 \quad (11)$$

```
#include <stdio.h>

int main() {
    int n = 3; // 3x3 matrix
    int complex_entries = n * n; // total complex entries

    // Step 1: Dimension of V (all complex entries over R)
    int dimV = complex_entries * 2; // 2 real numbers per complex
    entry

    // Step 2: Dimension of W1 (Hermitian)
    // Diagonal entries: n real numbers
    // Off-diagonal entries:  $n*(n-1)/2$  complex numbers =  $(n*(n-1))$ 
    // real
    int dimW1 = n + (n*(n-1)); // 3 + 6 = 9
}
```

```
// Step 3: Dimension of W2 (trace zero)
// Trace=0 gives 1 complex constraint = 2 real
int dimW2 = dimV - 2; // 18 - 2 = 16

// Step 4: Dimension of W1  W2
int dimIntersection = dimW1 - 1; // 1 real reduction from
    trace constraint

// Step 5: Dimension of W1 + W2
int dimSum = dimW1 + dimW2 - dimIntersection;

printf("%d\n", dimSum);

return 0;
}
```

```
import numpy as np

def dimension_W1_plus_W2(n=3):
    # Dimension of V: all  $n \times n$  complex matrices over  $\mathbb{R}$ 
    dimV = n * n * 2 # 2 real numbers per complex entry

    # Dimension of W1 (Hermitian)
    # Diagonal: n real numbers
    # Off-diagonal:  $n*(n-1)/2$  complex numbers =  $n*(n-1)$  real
    # numbers
    dimW1 = n + n*(n-1)

    # Dimension of W2 (trace zero)
    # Trace=0 gives 1 complex constraint = 2 real
    dimW2 = dimV - 2
```

```
# Dimension of W1  W2
dimIntersection = dimW1 - 1 # trace constraint reduces 1 real
    dimension

# Dimension of W1 + W2
dimSum = dimW1 + dimW2 - dimIntersection
return dimSum

if __name__ == __main__:
    n = 3 # 3x3 matrix
    print(dimension_W1_plus_W2(n))
```